

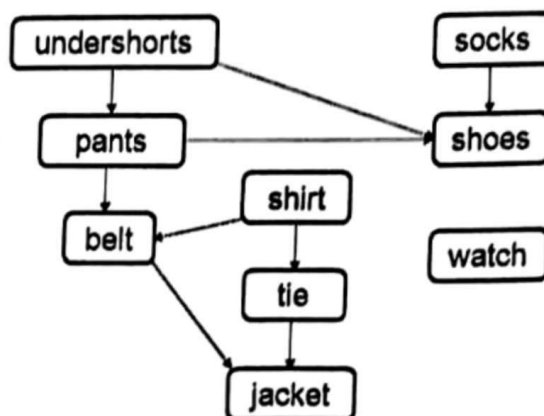
Time: 1 hour 20 minutes

Note: Answer all questions. Figure in the right margin indicate marks.

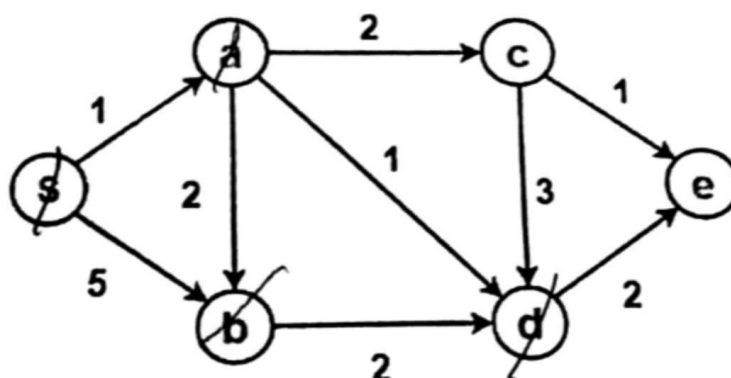
1. Draw a binary search tree using the following numbers. Also, show the linked representation of the BST. 6
 37, 2, 45, 41, 29, 20, 30, 40, 70
2. What is complete binary tree? Find the Huffman tree of the following 7
 information.

Letter	Freq
e	120
d	42
l	42
u	37
c	32
m	24
k	7
z	2

3. What is the minimum spanning tree? Find the discovery and finishing 7
 time of every node in following graph. Also find the forward and cross edge:



4. Explain the condition under which the Bellman-Ford algorithm cannot 10
 determine shortest paths. Find the single source shortest path using
 Dijkstra Algorithm. Assume source is S.



East West University

Department of Computer Science and Engineering

Title: Data Structures

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Fall 2025

Section: 10

Marks:30

Time: 1 hour 20 minutes

Note: Answer all questions. Figure in the right margin indicate marks.

1. Write a simple procedure to search an element into a sorted linked list. CO1-4
2. Mention the application of stack. Convert the following expression to equivalent postfix expression using stack: 9/CO2-10

$$A+B*(C^D-E)^{(F+G*H)}-I$$

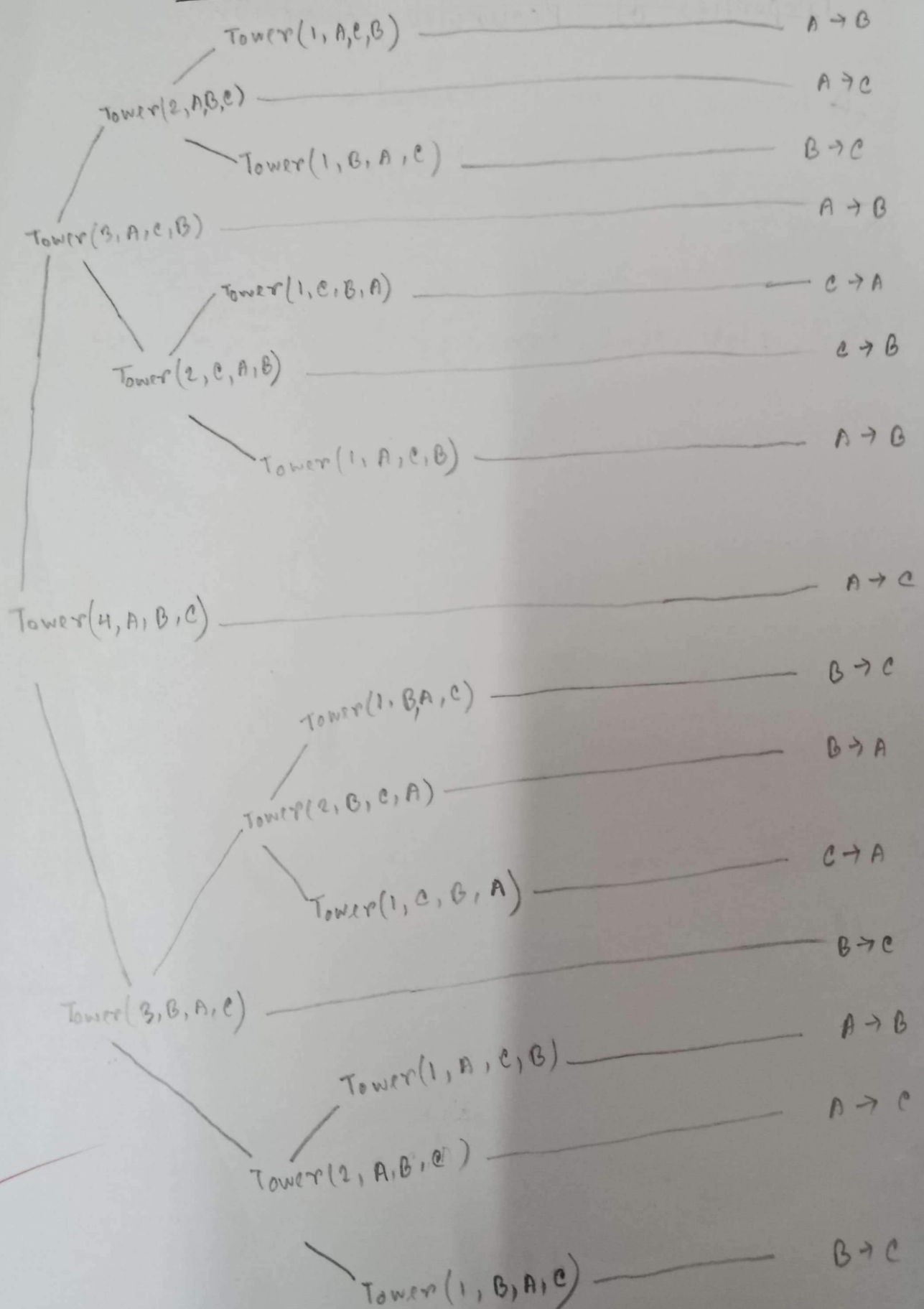
$$AB+CDE-{}^AFGH+{}^I-AF$$
3. Compare how linked list can be more useful for item insertion and deletion operation than an array. 3/CO1-6
4. What are the properties of recursive function? What are the properties of recursive function? State the solution of tower of Hanoi problem for $N=4$. 4/CO2-5
5. Suppose you have a circular queue of size 6 (1 to 6). Initially the value of the front and rear is 0. The current state of the queue is shown in the figure where the front=4 and the rear=6. Now show the state of the queue if the following operation is performed on the queue. Also find the value of front and rear in each step. 5/CO1-5

1	2	3	4	5	6
			12	72	35
			Front=4		Rear=6

insert (10); insert (5), delete(), delete(), insert (9); insert (9); insert (17); insert (15); insert (18); insert (11); insert (18).

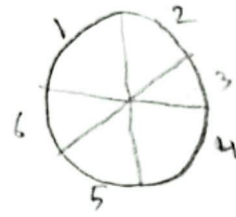
$$\frac{A \overline{B} \overline{C} B A A \overline{C}}{A \overline{B} \overline{C} / C \overline{A} / A \overline{B}}$$

Ans: To The Ques: NO - 4



Ans: To The Ques: No-5

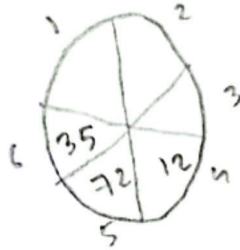
Initially $\text{front} = \text{rear} = 0$



Current state,

$\text{front} = 4$

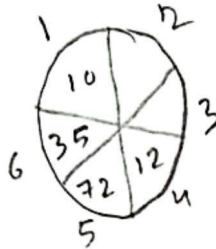
$\text{rear} = 6$



Insert (10),

$\text{front} = 4$

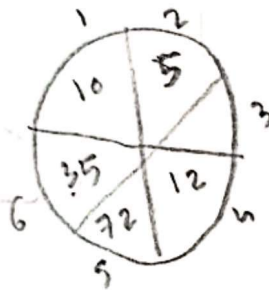
$\text{Rear} = 1$



Insert (5),

$\text{front} = 4$

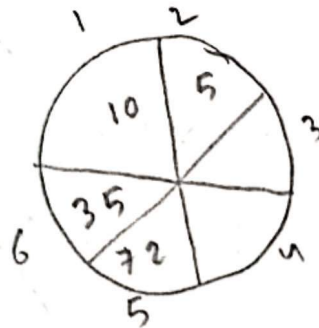
$\text{Rear} = 2$



~~de~~ delete ()

$\text{front} = 5$

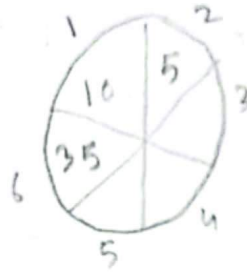
$\text{Rear} = 2$



delete (),

front = 6

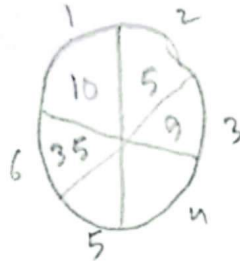
Rear = 2



Insert (9)

front = 6

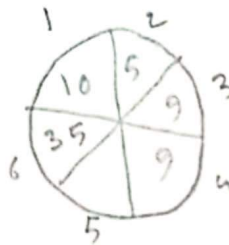
Rear = 3



Insert (9)

front = 6

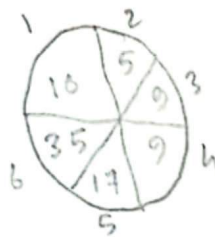
Rear = 4



Insert (17)

front = 6

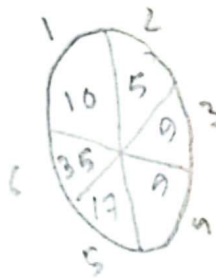
Rear = 5



Insert (15)

front = 6

Rear = 5



As front = Rear + 1, overflow

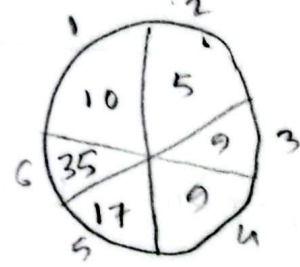
item can not be inserted

Insert (18)

front = 6

Rear = 5

As $\text{front} = \text{Rear} + 1$, overflow
item can not be inserted

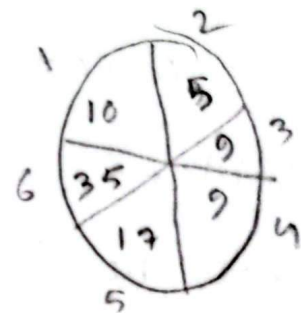


Insert (11)

front = 6

Rear = 5

As $\text{front} = \text{Rear} + 1$, overflow
item can not be inserted



Insert (18)

front = 6

Rear = 5

As $\text{front} = \text{Rear} + 1$, overflow
item can not be inserted

